

Name:

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TRANSPORT PHENOMENA I, FALL 2025

EXAM 3

Closed Book (Refer to the provided equations sheet)

Time: 90 minutes

1. A long vertical cylinder with radius R is rotating with constant angular velocity Ω within a large body of fluid.

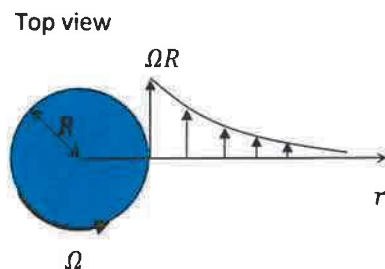
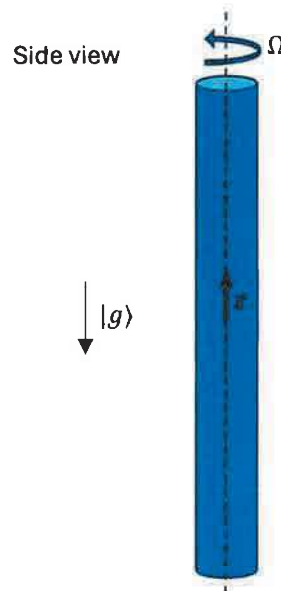
Postulates:

- Steady state, laminar flow.
- Newtonian fluid with constant viscosity μ and constant density ρ .
- Body of fluid around the cylinder is infinitely large.
- $v_r = 0$ and $v_z = 0$
- Pressure within the fluid is not a function of θ .

Answer the following questions in terms of parameters: ρ , μ , Ω , and R .

- 1.1. Using the mass balance (equation of continuity), demonstrate that v_θ is not a function of θ .

- 1.2. Using the equation of motion, derive an ordinary differential equation (ODE) for v_θ . Integrate the ODE to get an equation for v_θ as a function of r . Write down two boundary conditions for v_θ and evaluate the integration constants.



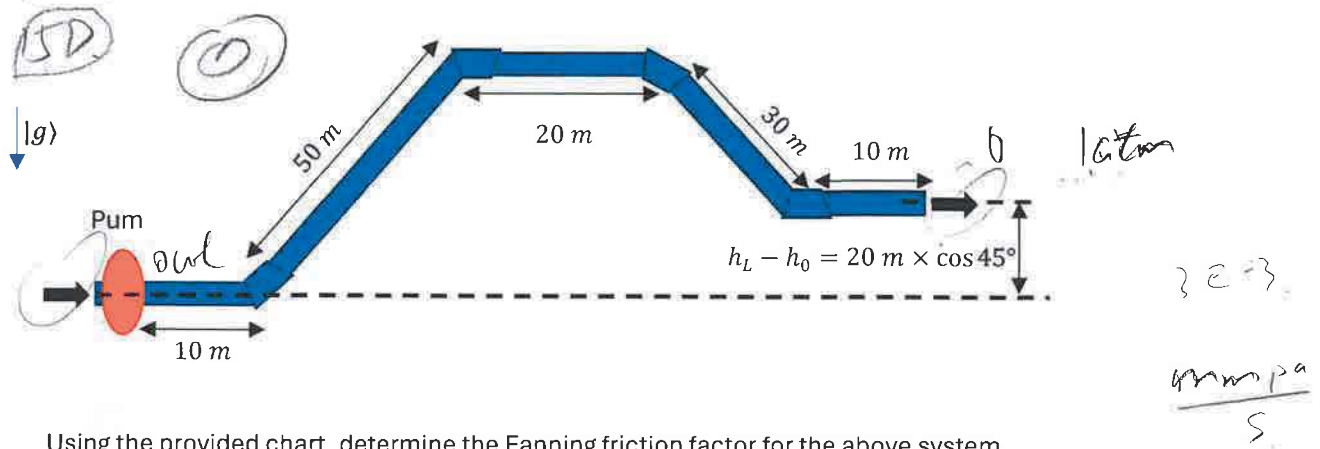
$$v_r = 0$$

$$v_{r=R} = \Omega R$$

$$v_z = 0$$

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2. Water is pumped between two reservoirs at the rate of $10 \frac{\text{kg}}{\text{s}}$. The pipeline consists of a pump, five smooth circular pipes of 20 cm inner radius (lengths of which are provided in the schematic below), and four 45° elbows. The resistance of each elbow is roughly equivalent to that offered by a pipe whose length is 15 diameters. The density of water is $\rho = 1000 \frac{\text{kg}}{\text{m}^3}$ and its viscosity is $\mu = 1.06 \text{ mPa} \cdot \text{s}$.



- 2.1. Using the provided chart, determine the Fanning friction factor for the above system.
- 2.2. If both reservoirs are at atmospheric pressure ($1.01 \times 10^5 \text{ Pa}$), what is the pressure at the pump outlet?

$$Re = \frac{\rho \langle v \rangle D}{\mu}$$

$$F_k \equiv (2\pi RL) \left(\frac{1}{2} \rho \langle v \rangle^2 \right) f = (P_0 - P_L) \pi R^2$$

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$$1.1 \quad \frac{\partial p}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (p r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (p v_\theta) + \frac{\partial}{\partial z} (p v_z) = 0$$

$$\frac{\partial p}{\partial t} + \frac{1}{r} \frac{\partial}{\partial \theta} (p v_\theta) = 0$$

Steady state $\frac{\partial p}{\partial t} = 0$

$$\frac{1}{r} \frac{\partial}{\partial \theta} (p v_\theta) = 0 \rightarrow \frac{1}{r} \neq 0 \rightarrow \frac{\partial}{\partial \theta} v_\theta = 0.$$

$$r \neq 0 \quad p \neq 0$$

$v_\theta = \text{constant}$ or
variables not θ .
are

So v_θ is not a function of θ

$$1.2 \quad p \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + v_z \frac{\partial v_\theta}{\partial z} + \frac{v_r v_\theta}{r} \right) = 0.$$

$$-\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r} \frac{\partial v_r}{\partial \theta} \right] + \frac{\partial p}{\partial \theta} = 0$$

$$\mu \left(\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) \right) = 0$$

$$\int \frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) = 0$$

$$\frac{\partial}{\partial r} (r v_\theta) = C_1 r$$

$$\int \frac{\partial}{\partial r} (r v_\theta) = \int C_1 r dr$$

$$r v_\theta = C_1 \frac{r^2}{2} + C_2 \quad \frac{1}{r}$$

$$v_\theta = C_1 \frac{r^2}{2} \frac{1}{r} + \frac{C_2}{r}$$

$$v_\theta = C_1 \frac{r}{2} + \frac{C_2}{r}$$

$$\begin{cases} v(r=A) = \Omega R \\ v(r \rightarrow \infty) = 0 \end{cases}$$

$$r = R = \Omega R = \frac{C_1 R}{2} + \frac{C_2}{R}$$

$$\lim_{r \rightarrow \infty} = 0 = \frac{C_1 r}{2} + \frac{C_2}{r}$$

$$\lim_{r \rightarrow \infty} \frac{C_1 r}{2} = 0 \quad C_1 = 0$$

$$\lim_{r \rightarrow \infty} \frac{C_2}{r} = 0$$

$$\Omega R = \frac{C_2}{R}$$

$$\Omega R^2 = C_2$$

$$\begin{cases} C_1 = 0 \\ C_2 = \Omega R^2 \end{cases}$$

$$v_\theta = 0 \frac{r}{2} + \frac{\Omega R^2}{r}$$

$$v_\theta = \frac{\Omega R^2}{r}$$

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$$2.1 \quad \omega = 10 \frac{\text{kg}}{\text{s}}$$

$$P_0 = 1.01 \times 10^5 \text{ Pa}$$

$$r = 20 \text{ cm} \quad D = 40 \text{ cm} = 0.4 \text{ m}$$

$$L = 10 + 50 + 20 + 30 + 10 + 15D + 15D + 15D + 15D$$

$$= 120 + 24$$

$$= 144 \text{ m}$$

$$\rho = 1000 \frac{\text{kg}}{\text{m}^3}$$

$$\mu = 1.06 \text{ mPa} \cdot \text{s}$$

$$\omega = \pi r^2 \rho v$$

$$\frac{\frac{\text{kg}}{\text{s}}}{\pi \text{ m}^2 \frac{\text{kg}}{\text{m}^3}}$$

$$v = \frac{\omega}{\pi r^2 \rho} = \frac{10 \frac{\text{kg}}{\text{s}}}{\pi (0.2)^2 \cdot 1000 \frac{\text{kg}}{\text{m}^3}} = 0.0796 = 0.08 \frac{\text{m}}{\text{s}}$$

$$Re = \frac{1000 \times 0.08 \times 0.4}{1.06 \times 10^{-3}} = 30188.67 > 2100 \text{ turbulent}$$

$$= 3.02 \times 10^4 \checkmark$$

$$By \text{ chart 6.22 } f \approx 0.006 \checkmark$$

$$2.2 \quad F_k = \pi R L \frac{1}{2} \rho v^2 f = (P_0 - P_L) \pi R^2$$

$$\frac{\pi R L \rho v^2 f}{\pi R^2} = P_0 - P_L \rightarrow \frac{L \rho v^2 f}{R} = (P_0 - P_L) + \rho g (h_0 - h_L)$$

$$\frac{L \rho v^2 f}{R} - \rho g (h_0 - h_L) = P_0 - P_L$$

$$\frac{L \rho v^2 f}{R} + \rho g (h_L - h_0) = P_0 - P_L$$

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$$\frac{144m \times 1000 \frac{kg}{m^3} \times (0.08 \frac{m^2}{s^2})^2 \times 0.006}{0.2m} - 1000 \frac{kg}{m^3} \times 9.8 \frac{m}{s^2} (h_0 - h_L)$$

$$27.648 - 9800 \frac{kg}{m^2 s^2} (h_0 - h_L)$$

$20m \times \cos 45^\circ m$

$$27.648 \frac{kg}{m s^2} - 9800 \frac{kg}{m s^2} \times (-10\sqrt{2} m)$$

$$= 138620.5771 \frac{kg}{m s^2} = P_0 - P_L$$

$$138620.5771 \frac{kg}{m s^2} = P_0 - 1.01 \times 10^5 \frac{kg}{m s^2}$$

$$138620.5771 Pa + 1.01 \times 10^5 Pa = P_0$$

$$239620.5771 Pa = P_0 \checkmark$$

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